

Rezdiffra (Resmetirom)

The First FDA-Approved Pharmacotherapy for Metabolic-Associated Steatohepatitis (MASH)

A Literature Review | Drug Discovery · Clinical Trials · Therapeutic Applications · Market Impact

Abstract. Resmetirom (brand name: Rezdiffra), developed by Madrigal Pharmaceuticals, Inc., represents a landmark achievement in hepatology. On March 14, 2024, the U.S. Food and Drug Administration (FDA) granted accelerated approval to resmetirom, making it the first and only pharmacological treatment for noncirrhotic nonalcoholic steatohepatitis (NASH) with moderate to advanced liver fibrosis (F2–F3). This literature review traces the drug's journey from molecular design to regulatory approval, covering its mechanism of action as a liver-directed thyroid hormone receptor-beta (THR- β) selective agonist, the 18-study clinical development program culminating in the pivotal Phase 3 MAESTRO-NASH trial, its therapeutic profile and safety data, and the substantial commercial and scientific impact it has had on the MASH treatment landscape.

1. Introduction

Nonalcoholic fatty liver disease (NAFLD) and its progressive inflammatory form, nonalcoholic steatohepatitis (NASH), have emerged as leading causes of chronic liver disease worldwide. NASH is now increasingly referred to by its mechanistically accurate name, metabolic dysfunction-associated steatohepatitis (MASH). The disease is characterised by hepatic steatosis, lobular inflammation, and hepatocyte ballooning, which can progress to fibrosis, cirrhosis, and ultimately liver failure or hepatocellular carcinoma. Globally, an estimated 115 million people have NASH, with 6 to 8 million Americans alone carrying moderate-to-advanced fibrosis — the subgroup at greatest risk of clinical deterioration.

Despite decades of research, no pharmacotherapy existed for NASH until 2024. Patients were limited to lifestyle modification: caloric restriction and increased physical activity. Multiple late-stage candidates — including obeticholic acid, cenicriviroc, and selonsertib — failed to demonstrate the required combination of NASH resolution and fibrosis improvement in Phase 3 trials, illustrating the difficulty of the therapeutic target. Against this backdrop, resmetirom's approval represents not merely a new drug but the opening of an entirely new therapeutic category.

2. Drug Discovery Process

2.1 Biological Rationale and Target Identification

The scientific basis for resmetirom originates in the role of thyroid hormone in hepatic lipid metabolism. Thyroid hormone exerts its effects through two nuclear receptor isoforms: THR-alpha, expressed predominantly in cardiac and skeletal muscle tissue, and THR-beta, expressed primarily in the liver, brain, and pituitary gland. Activation of hepatic THR-beta drives expression of genes involved in fatty acid oxidation, lipogenesis inhibition, and lipoprotein clearance — all processes directly implicated in

NASH pathogenesis.

Historical attempts to exploit the thyroid hormone axis for metabolic disease were derailed by the non-selective activation of THR- α , which produced adverse cardiac effects including arrhythmia and tachycardia, as well as bone loss. Earlier compounds such as eprotirome and MGL-3196 precursors showed that receptor subtype selectivity and hepatic targeting were prerequisites for a viable drug. The key insight from Madrigal founder Dr. Becky Taub and her team, developed over more than 15 years, was that a truly beta-selective, liver-directed small molecule could unlock the metabolic benefits of THR activation while avoiding systemic off-target toxicities.

2.2 Molecular Design and Optimisation

Resmetirom (chemical name: 2-[3,5-dichloro-4-[(5-isopropyl-6-oxo-1,6-dihydropyridazin-3-yl)oxy]phenyl]-3,5-dioxo-2,3,4,5-tetrahydro-1,2,4-triazine-6-carbonitrile) is an orally administered small molecule with a molecular weight of approximately 491 Da. The compound was engineered to achieve three key pharmacological properties: (1) high selectivity for THR- β over THR- α , thereby reducing cardiac risk; (2) preferential uptake by hepatocytes via organic anion transporting polypeptide (OATP) transporters, achieving liver-directed concentrations; and (3) minimal systemic distribution, preventing activation of THR- β in the pituitary that could suppress thyroid-stimulating hormone (TSH) and cause hyperthyroidism.

Preclinical studies in rodent and non-human primate models confirmed that resmetirom reduced hepatic triglycerides, improved histological markers of steatohepatitis, and reduced serum LDL-cholesterol — consistent with THR- β -mediated enhancement of reverse cholesterol transport. Critically, at efficacious doses, the compound did not affect heart rate, bone density, or thyroid hormone axis parameters, supporting advancement into human trials.

3. Clinical Trial Phases

3.1 Phase 1 Studies

The clinical development program for resmetirom encompassed 18 studies in total: 12 Phase 1 studies, 2 Phase 2 studies, and 4 Phase 3 studies. The Phase 1 programme evaluated single- and multiple-ascending doses in healthy volunteers, characterising the pharmacokinetic profile, safety, tolerability, and drug-drug interaction potential. Results confirmed the liver-directed distribution hypothesis: plasma concentrations were low relative to estimated hepatic concentrations, and no clinically meaningful changes in heart rate, TSH, or bone turnover markers were observed. The half-life supported once-daily oral dosing, and hepatic enzyme inhibition studies informed the drug interaction profile.

3.2 Phase 2 Studies

Two Phase 2 trials established proof-of-concept in patients with biopsy-confirmed NASH. The first, a 12-week randomised controlled study published in 2019 in the *New England Journal of Medicine*, demonstrated that resmetirom significantly reduced hepatic fat content measured by MRI-proton density fat fraction (MRI-PDFF) compared with placebo (36.3% relative reduction vs. 9.3%; $p < 0.001$).

Liver enzyme levels (ALT, AST) and atherogenic lipid markers including LDL-cholesterol and triglycerides were also significantly reduced. These findings provided the mechanistic rationale for advancing to a histology-based Phase 3 programme.

A longer 36-week Phase 2 extension published in 2020 confirmed durability of hepatic fat reduction and added preliminary evidence of histological improvement on liver biopsy. The favourable safety profile across these studies — with mild-to-moderate diarrhoea and nausea as the most common adverse events — supported the risk-benefit profile required for a Phase 3 submission.

3.3 Phase 3 Trials: The MAESTRO Programme

The Phase 3 programme, called MAESTRO (Multiple Ascending doses Evaluating STEatohepatitis Reduction Outcomes), comprised four trials. The pivotal study was MAESTRO-NASH, a multicentre, randomised, double-blind, placebo-controlled trial that enrolled 1,759 adults with biopsy-confirmed NASH and F1–F3 fibrosis. Patients were randomised 1:1:1 to resmetirom 80 mg, resmetirom 100 mg, or placebo, administered orally once daily for 52 weeks. The FDA pre-specified dual histological primary endpoints were: (1) NASH resolution with no worsening of fibrosis, and (2) improvement of at least one fibrosis stage with no worsening of NASH Activity Score (NAS).

Results published in the *New England Journal of Medicine* in February 2024 demonstrated that both primary endpoints were met. NASH resolution was achieved by 25.9% (80 mg) and 29.9% (100 mg) of resmetirom-treated patients versus 9.7% in the placebo group (both $p < 0.001$). Fibrosis improvement of at least one stage was achieved in 24.2% (80 mg) and 25.9% (100 mg) versus 14.2% with placebo (both $p < 0.001$). Additionally, 91% of patients treated with resmetirom 100 mg achieved at least one primary endpoint. Significant reductions in LDL-cholesterol and non-HDL-cholesterol were also observed, suggesting potential cardiovascular benefits. Serious adverse event rates were comparable across all groups (approximately 11–13%), further supporting the drug's safety.

Three complementary MAESTRO trials evaluated resmetirom in NAFLD without mandated biopsy (MAESTRO-NAFLD-1), as well as providing long-term safety and outcomes data. MAESTRO-NASH continues as an ongoing outcomes study, with participants treated for up to 54 months to capture hard clinical endpoints such as progression to cirrhosis and hepatic decompensation — data which, if positive, will support conversion from accelerated to full FDA approval.

3.4 Regulatory Pathway

Resmetirom benefited from multiple FDA expedited programmes. It received Breakthrough Therapy Designation in April 2023, which enabled intensive FDA guidance during development. The New Drug Application (NDA) was submitted in July 2023 and accepted with Priority Review designation in September 2023. Accelerated approval on March 14, 2024 was based on the histological endpoints of NASH resolution and fibrosis improvement, which the FDA deemed reasonably likely to predict clinical benefit.

4. Therapeutic Applications

4.1 Approved Indication

Resmetirom is approved for use in conjunction with diet and exercise for the treatment of adults with noncirrhotic NASH (MASH) with moderate to advanced liver fibrosis, defined as stages F2 to F3 on the METAVIR or equivalent histological scoring systems. The drug is administered as a once-daily oral tablet: 80 mg for patients weighing less than 100 kg and 100 mg for patients at or above 100 kg. Patients do not necessarily require a liver biopsy for diagnosis; the prescribing label specifies that less than 5% of commercially covered patients require biopsy, as non-invasive tests and imaging are increasingly accepted for diagnosis in clinical practice.

4.2 Mechanism in the Disease Context

In MASH, lower THR-beta activity in the liver is predictive of hepatic decompensation. By selectively agonising THR-beta, resmetirom activates mitochondrial fatty acid beta-oxidation pathways and suppresses de novo lipogenesis, directly addressing the lipotoxicity that drives hepatocyte injury and inflammation. Downstream effects include reduced reactive oxygen species generation, decreased inflammatory cytokine signalling, and attenuation of hepatic stellate cell activation — the cellular process underlying fibrogenesis. The LDL-lowering effect (15–20% reduction in trials) additionally provides cardiovascular risk mitigation in a patient population already burdened by metabolic comorbidities such as type 2 diabetes and hypertension.

4.3 Emerging and Investigational Uses

The MAESTRO-NASH OUTCOMES trial, which completed enrolment in October 2024, is evaluating resmetirom in compensated MASH cirrhosis (F4c). Positive outcomes data, expected in approximately 2027, could substantially expand the approved patient population and position resmetirom as the first therapy spanning the full F2–F4 fibrosis spectrum. Preliminary two-year data from MAESTRO-NAFLD-1 showed that patients with compensated cirrhosis achieved a mean 6.7 kPa reduction in liver stiffness, with 51% achieving at least a 25% reduction — early signals of meaningful benefit in this harder-to-treat group. The drug's LDL-lowering properties have also raised interest in potential applications in patients with metabolic syndrome and cardiovascular disease.

4.4 Safety Profile Summary

The most frequently reported adverse effects in clinical trials were gastrointestinal: diarrhoea (25–33%) and nausea (14–19%), predominantly mild-to-moderate in severity and resolving without dose discontinuation in most cases. The incidence of serious adverse events was similar between resmetirom and placebo groups. No clinically meaningful effects on heart rate, thyroid function tests, TSH levels, or bone mineral density were observed, confirming the selectivity and hepatic-targeting design. Drug interactions via CYP3A4 and OATP transporters require monitoring in patients on concomitant statin therapy, as resmetirom may increase statin exposure.

5. Market Impact

5.1 Commercial Launch and Uptake

Rezdiffra was commercially launched in the United States in April 2024, approximately three weeks after FDA approval. The launch was one of the fastest in hepatology, reflecting years of unmet need

and high prescriber awareness. The annual list price was set at \$47,400, which aligned with independent health-technology assessment benchmarks for cost-effectiveness in the target population. Formulary access expanded rapidly: by the third quarter of 2024, more than 80% of commercial lives were covered by plans including Rezdiffra, with fewer than 5% of covered lives requiring liver biopsy for prior authorisation — a significant reduction in access barriers compared with earlier projections.

5.2 Financial Performance

Rezdiffra generated net sales of \$180.1 million in its first full year of commercialisation (2024), with Q4 alone contributing \$103.3 million — reflecting accelerating uptake. By year-end 2024, more than 11,800 patients were on therapy. Madrigal Pharmaceuticals closed the year with approximately \$931 million in cash and marketable securities, providing a substantial runway for the ongoing Phase 3 outcomes trial and the planned European launch. These figures represent a strong commercial debut for a drug entering a previously untreated disease, though the patient population under active specialist care in the U.S. (~315,000 with F2–F3 fibrosis) signals substantial headroom for continued growth.

5.3 Scientific and Clinical Practice Impact

Rezdiffra's approval has catalysed a restructuring of MASH clinical practice guidelines. Both the American Association for the Study of Liver Diseases (AASLD) and the European Association for the Study of the Liver (EASL) updated their guidance in 2024 to include resmetirom as a first-line pharmacological option for eligible patients. The MAESTRO-NASH Phase 3 paper in the *New England Journal of Medicine* was selected as one of the journal's 14 notable scientific articles of 2024, underscoring its scientific significance. The success of resmetirom has also reinvigorated pharmaceutical investment in the MASH space after a prolonged period of failures; several combination regimen trials pairing resmetirom with GLP-1 receptor agonists, FXR agonists, and ACC inhibitors are now underway, reflecting optimism that resmetirom's proof-of-concept can serve as a backbone for next-generation MASH therapy.

5.4 Global Expansion

Madrigal submitted a Marketing Authorisation Application (MAA) to the European Medicines Agency (EMA) in March 2024. The EMA's Committee for Medicinal Products for Human Use (CHMP) opinion was anticipated in mid-2025, with Madrigal planning country-by-country commercial launches beginning with Germany in the second half of 2025. A European approval would position Rezdiffra as the first MASH-specific therapy on that continent. Additional regulatory filings in Asia-Pacific markets are expected as the global MASH burden continues to rise alongside increasing rates of obesity and metabolic syndrome.

6. Conclusion

Resmetirom (Rezdiffra) stands as one of the most consequential drug approvals of the 2020s. Its development exemplifies the power of targeted molecular design — engineering receptor selectivity and tissue targeting to unlock the therapeutic potential of a well-understood biological axis while eliminating the toxicities that had frustrated earlier attempts. The comprehensive 18-study clinical programme, culminating in the dual-endpoint MAESTRO-NASH Phase 3 trial, provided robust

histological evidence in a disease where histology has been the FDA's gold standard. The commercial launch has validated market demand, and the pipeline of combination trials and indication expansions suggests resmetirom's role in MASH management will only grow. For a disease affecting hundreds of millions globally and with no prior pharmacological option, this approval represents a genuine turning point.

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